

Erik Jonsson School of Engineering and Computer Science

Department of Mechanical Engineering

Overview

The objective of the Bachelor of Science degree program in Mechanical Engineering is to produce Mechanical Engineering graduates who will be capable of undertaking challenging projects that require knowledge of the fundamentals and design of mechanical and thermal systems. The program seeks to build Mechanical Engineers to meet the needs of analysis, design, and development in industry, as well as to educate them to be innovators and policy makers. The BS degree program will provide the necessary training and education for future engineers who will effectively identify new problems and develop innovative solutions, including new manufacturing and fabrication technologies.

Mechanical Engineering (BS)

Program Educational Objectives for Mechanical Engineering

One broad goal for the Erik Jonsson School is to provide an excellent education for our students. Within a few years after graduation, graduates of the Mechanical Engineering Program should:

- Be on a successful career path as competent professionals in their chosen field or by pursuing advanced study.
- Serve their professional roles in a responsible and ethical manner to meet the needs of engineering and society.
- Be effective contributors or leaders in professional settings, including contributions to multidisciplinary teams.
- Actively pursue life-long learning through advanced education, self-study, professional development, etc.

High School Preparation

Engineering education requires a strong high school preparation. Pre-engineering students should have high school preparation of at least one-half year in trigonometry and at least one year in elementary algebra, intermediate and advanced algebra, geometry, pre-calculus, chemistry, and

physics, thus developing their competencies to the highest possible levels and preparing to move immediately into demanding college courses in calculus, calculus-based physics, and chemistry for science majors. It is also essential that pre-engineering students have the competence of reading comprehension, and to write logically, clearly and correctly.

ABET Accreditation

The BS program in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET, www.abet.org.

Lower-Division Study

All lower-division students in Mechanical Engineering concentrate on mathematics, science, and introductory engineering courses, building competence in these cornerstone areas for future application in upper-division engineering courses. The following requirements apply both to students seeking to transfer to UT Dallas from other institutions as well as to those currently enrolled at UT Dallas, whether in another school or in the Erik Jonsson School of Engineering and Computer Science.

Academic Progress in Mechanical Engineering

In order to make satisfactory academic progress as a Mechanical Engineering major, a student must meet all University requirements for academic progress, and must earn a grade of C- or better in each of the "major requirements" courses. No "Major Requirements" course may be taken until the student has obtained a grade of C- or better in each of the prerequisites. If a higher grade requirement is stated for a specific class, the higher requirement applies.

Bachelor of Science in Mechanical Engineering

[Degree Requirements](#) (127 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: William Anderson, Nandika D'Souza, D. Todd Griffith, Fatemeh Hassanipour, Stefano Leonardi, Yaoyu Li, Hongbing Lu, Arif Malik, Majid Minary-Jolandan, Dong Qian, Zhenpeng Qin, Mario A. Rotea, Joshua Summers, Yonas Tadesse, Edward White, Seung M. You, Jie Zhang

Associate Professors: Rodrigo Bernal Montoya, Giacomo (Valerio) Iungo, Xinfang Jin, Justin Koeln, Golden Kumar, Kristin Miller, Justin Ruths, Tyler Summers, Yue Zhou

Assistant Professors: Connor Armstrong, Shuang (Cynthia) Cui, Yaqing Jin, Justin Koeln, Juyoung Leem, Jiannan Li, Wei Li, Jen Liu, Xinchun Ni, Hui Ouyang, Zejiang Wang, Guoping Xiong, Yanwen Xu, Kianoosh Yousefi, Armin Zare

Associate Professors of Instruction: Wooram Park

UT Dallas Affiliated Faculty: Waseem Abbas, Kyeongjae (KJ) Cho, Babak Fahimi, Matthew J. Goeckner, Jiyoung Kim, Moon J. Kim, S.O. Reza Moheimani, Mehrdad Nourani, Linda Smith-Brecheisen, Mark W. Spong, Robert M. Wallace, Steve Yurkovich, Jie Zheng

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

[RHET 1302](#) College Writing II¹

[ECS 2390](#) Professional and Technical Communication¹

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I^{2, 3}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics⁴

[PHYS 2326](#) Electromagnetism and Waves⁴

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Option](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2417](#) Calculus I ^{2, 3}

[MATH 2419](#) Calculus II ^{2, 3}

[PHYS 2125](#) Physics Laboratory I ⁴

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 82 semester credit hours ⁵

Major Preparatory Courses: 29 semester credit hours beyond Core Curriculum

Students pursuing the general program take 29 semester credit hours from the list below:

[CHEM 1111](#) General Chemistry Laboratory I

[CHEM 1311](#) General Chemistry I

[ECS 1100](#) Introduction to Engineering and Computer Science

[ECS 1210](#) Foundations in Engineering and Computer Science I

[ECS 1220](#) Foundations in Engineering and Computer Science II

[MATH 2417](#) Calculus I ^{2, 3}

[MATH 2419](#) Calculus II^{2, 3}

[MATH 2420](#) Differential Equations with Applications

[MECH 1208](#) Introduction to Mechanical Engineering⁶

[MECH 2120](#) Mechanics of Materials Laboratory

[ENGR 2300](#) Linear Algebra for Engineers

[MECH 2310](#) Statics

[MECH 2320](#) Mechanics of Materials

[MECH 2330](#) Dynamics

[PHYS 2125](#) Physics Laboratory I⁴

[PHYS 2126](#) Physics Laboratory II

[PHYS 2325](#) Mechanics⁴

[PHYS 2326](#) Electromagnetism and Waves⁴

[RHET 1302](#) College Writing II¹

Major Core Courses: 41 semester credit hours beyond Core Curriculum

[ECS 2390](#) Professional and Technical Communication¹

[MECH 3105](#) Computer Aided Design Laboratory

[MECH 3115](#) Fluid Mechanics Laboratory

[MECH 3120](#) Heat Transfer Laboratory

[MECH 3150](#) Kinematics and Dynamics Laboratory

[MECH 3300](#) Applied Engineering Mathematics

[MECH 3305](#) Computer Aided Design

[MECH 3310](#) Thermodynamics

[MECH 3315](#) Fluid Mechanics

[MECH 3320](#) Heat Transfer

[MECH 3340](#) System Dynamics Modeling and Analysis

[ENGR 3341](#) Probability Theory and Statistics

[MECH 3350](#) Kinematics and Dynamics of Mechanical Systems

[MECH 3351](#) Design of Mechanical Systems

[MECH 4110](#) Systems and Controls Laboratory

[MECH 4310](#) Systems and Controls

[MECH 4381](#) Senior Design Project I ⁷

[MECH 4382](#) Senior Design Project II ⁷

Technical Electives: 12 semester credit hours

A student is required to take at least 9 semester credit hours of courses from the following list. The student can take the remaining semester credit hours from 3000 or higher level ECS courses upon approval from the department.

[MECH 2340](#) Circuits and Applied Electronics

[MECH 3360](#) Materials Science and Engineering

[MECH 3370](#) Applied Thermodynamics

[MECH 3380](#) Computational Design and Analysis

[MECH 3381](#) Fundamental Manufacturing Processes

[MECH 3382](#) Additive Manufacturing

[MECH 3V95](#) Topics in Mechanical Engineering

[MECH 4301](#) Intermediate Mechanics of Materials

[MECH 4320](#) Applications of Computational Tools in Thermal Fluid Science

[MECH 4330](#) Intermediate Fluid Mechanics

[MECH 4340](#) Mechanical Vibrations

[MECH 4342](#) Fundamentals of Robotics

[MECH 4345](#) Soft Robotics

[MECH 4360](#) Fundamentals of Nanostructured Materials

[MECH 4365](#) Energy Analytics

[MECH 4370](#) Principles of MEMS

[MECH 4375](#) Design for Manufacturing and Assembly

[MECH 4380](#) HVAC Systems

[MECH 4387](#) Engineering Economy

[MECH 4V95](#) Topics in Mechanical Engineering

[ENTP 3301](#) Innovation and Entrepreneurship

III. Elective Requirements: 3 semester credit hours

Free Electives: 3 semester credit hours

Both lower- and upper-division courses may count as free electives.

Degree programs in the Erik Jonsson School of Engineering and Computer Science are governed by various accreditation boards that place restrictions on classes used to meet the curricular requirements of degrees they certify. For this reason, not all classes offered by the University can be used to meet elective requirements. Please check with your academic advisor before enrolling in classes you hope to use as free electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Fast Track Baccalaureate/Master's Degrees

In response to the need for advanced education in Mechanical Engineering, a Fast Track program is available to well-qualified UT Dallas undergraduate students. Qualified seniors may take up to 15 graduate semester credit hours that may be used to complete the baccalaureate degree and also to satisfy the requirements for the master's degree. This is accomplished by (1) taking courses (typically electives) during one or more summer semesters, and (2) beginning graduate coursework during the senior year. Details are available from the Associate Dean for Undergraduate Education.

Honors Program

The Department of Mechanical Engineering offers upper-division Honors for outstanding students in the BS Mechanical Engineering degree program. This program offers special sections of designated classes and other activities designed to enhance the educational experience of

exceptional students. Admission to the Honors programs requires a 3.500 or better GPA (grade point average) in at least 30 semester credit hours of coursework. Graduation with Honors requires a 3.500 or better GPA and completion of at least 6 honors classes. These honors classes must include either [MECH 4399](#) or [MECH 4V98](#) and a Senior Honors Thesis must be completed within one of those two classes. While the topics may be related, the Senior Thesis does not replace the need for the student to complete a regular Senior Design Project. The other 5 honors classes can come from a mixture of Graduate level (up to a count of 4) classes and special honor sections of regular undergraduate MECH classes (up to a count of 2).

Departmental Honors with Distinction may be awarded to students whose Senior Honors Thesis is judged by a faculty committee to be of exemplary quality. Only students graduating with Departmental Honors are eligible. Thesis/projects must be submitted by the deadline that applies to MS Theses in the graduating semester to allow for proper evaluation. Students interested in Honors with Distinction are encouraged to start working on their thesis/project a year prior to graduation.

Minors

The Department of Mechanical Engineering does not offer minors at this time.

1. Semester credit hours fulfill the communication component of the Core Curriculum.
2. Three semester credit hours of Calculus are counted under Mathematics Core, and five semester credit hours of Calculus are counted as Component Area Option Core.
3. MATH 2417 and MATH 2419 requirements can be fulfilled by completing MATH 2413, MATH 2414, and MATH 2415, as is commonly done when enrollment prerequisites for MATH 2417 are not met. Students who take MATH 2413, MATH 2414, and MATH 2415 may request three additional semester credit hours of Mathematics be applied to fulfill Free Elective credit requirements and may petition that one additional credit hour of Mathematics be applied to fulfill Technical Elective credit requirements.
4. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted as Component Area Option Core.
5. Students must pass each of the "Major Requirements" courses listed in this degree plan and each of their prerequisites, with a grade of C- or better.
6. Transfer students with sufficient background may petition to substitute upper-division semester credit hours in the major for this class.
7. MECH 4381 and 4382 must be taken in successive semesters (fall/spring, spring/summer, summer/fall) to complete a two-semester project.

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